

Understanding as Key

One-pager · Bilateral trust primitive for multi-intelligence coordination

A trust primitive for coordination across heterogeneous counterparty types — human, machine, ecological, institutional, and other. Replaces possession-based authentication (token, certificate, key) with comprehension-based attestation: a counterparty establishes trust by holding and rehydrating a bilateral compression jointly authored with a partner.

Protocol (five steps):

1. Shared vocabulary commitment (hashed, signed).
2. Substrate-path agreement on a UOR-mapped topology (graph, ontology, dataset).
3. Bilateral traversal in lockstep, lap-by-lap signed.
4. Independent natural-language compression by each party (P_A , P_B).
5. Bilateral witness with chain anchor of the commitment $x = \text{hash}(P_A \parallel P_B)$.

Inscription paths (visibility ratio $\sigma \in [0,1]$):

- **Symmetric ($\sigma = 0$):** only x published. Neither compression visible.
- **Asymmetric ($\sigma = 1$):** P_A published; P_B committed only in the hash.
- **Interleaved ($0 < \sigma < 1$):** fragments of both compressions interweave across the visibility boundary. Sweet spots at $\phi^{-1} = 0.382$ and $\phi = 0.618$.

Verification. Trust is established later by rehydration: the bearer unfolds their compression into a structure consistent with the chain-anchored commitment. Faithful unfolding succeeds; drift or substitution fails. No long-lived secret is stored. Verification is independent of session-key custody.

Multi-intelligence property. Substrate selection is neutral across counterparty type. The protocol admits human, machine, ecological, and other intelligences as parties to the same primitive.

Reference: sync.soulbis.com/p/understanding-as-key